



UCSC

University of Colombo, Sri Lanka

University of Colombo School of Computing



**DEGREE OF BACHELOR OF INFORMATION TECHNOLOGY
(EXTERNAL)**

Academic Year 2025 — 1st Year Examination — Semester 1

IT1206 — Computer Systems

Multiple Choice Question Paper
(2 Hours)

Important Instructions

- The duration of the paper is **2 Hours**.
- The medium of instructions and questions is English.
- This paper has **40 questions** on **11 pages**. Answer **all** questions.
- All questions are of the **MCQ** (Multiple Choice Questions) type.
- Each question will have **5 (five)** choices with **ONE OR MORE** correct answers.
- This paper consists of 100 marks and all the questions will carry equal marks.
- There will be a penalty for incorrect responses to discourage guessing.
- The mark given for a question will vary from -1 (All the incorrect choices are marked & no correct choices are marked) to +1 (All the correct choices are marked & no incorrect choices are marked). However, **the minimum mark per question would be zero**.
- Answers should be marked on the **special answer sheet** provided.
- Note that questions appear on both sides of the paper. If a page or part of a page is not printed, please inform the supervisor/invigator immediately.
- Mark the correct choices on the question paper first and then transfer them to the given answer sheet which will be machine marked. **Please completely read and follow the instructions given on the other side of the answer sheet before you shade your correct choices.**
- Any electronic device capable of storing and retrieving text, including electronic dictionaries, smartwatches, and mobile phones, is not allowed.
- Calculators are **not** allowed.
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- 1) Which of the following statements accurately describe limitations or issues inherent to Floating-Point arithmetic in computer systems?
- (a) Operations are prone to overflow and underflow,
(b) Operations can introduce rounding off error.
(c) Associativity is not guaranteed: e.g., $(a + b) + c$ may not equal $a + (b + c)$,
(d) Distributivity is not guaranteed: e.g., $a \times (b + c)$ may not equal $(a \times b) + (a \times c)$,
(e) Floating-point numbers need higher storage space than integer numbers.
- 2) Which of the following is/are **NOT** a part of a typical computer system's hardware organization?
- (a) Output (b) CPU (c) Throughput
(d) Storage (e) Latency
- 3) Which of the following number systems allow each digit to be directly converted into a fixed-length binary representation, and those representations in concatenation gives the correct binary value?
- (a) Octal (base 8) (b) Decimal (base 10) (c) Hexadecimal (base 16)
(d) Sexagesimal (base 60) (e) None of the Above
- 4) Which of the following devices can function as both input and output devices in a computer system?
- (a) Keyboard (b) Mouse (c) Modem
(d) Plotter (e) Touchscreen
- 5) Which of the following storage devices provides the **fastest random access** to data?
- (a) Hard Disk Drive (b) Solid State Drive (c) Floppy Disk Drive
(d) Optical Storages (e) Magnetic Tape
- 6) Which of the following terms is used to describe the smaller, but faster memory placed closest to the CPU to store frequently used instructions and data?
- (a) Main Memory (b) Register (c) Cache Memory
(d) Intensity Buffer (e) Rapid Access Memory
- 7) Which of the following expansion cards is/are correctly matched with its primary function?
- (a) Graphics Card – Manages audio output
(b) Sound Card – Processes audio signals
(c) Network Interface Card – Provides network connectivity
(d) TV Tuner Card – Increases network throughput
(e) USB Card – Manages storage requirements of external hard disks

8) Which of the following statements is/are correct?

- (a) Volatile memory retains its contents even when the power supply is turned off.
- (b) Cache memory is an example of non-volatile storage.
- (c) Main memory is volatile and loses its data when the power is switched off.
- (d) Hard disk drives are non-volatile storages and SSDs are volatile storages.
- (e) Non-volatile storage is primarily used for temporary data during program execution.

9) Which of the following statements correctly describe(s) a mechanical or operational aspect of a Hard Disk Drive?

- (a) It only uses sequential data access.
- (b) It uses spinning read/write head to read and write data.
- (c) It uses spinning magnetic platters to store data.
- (d) It uses laser to read and write data on platters.
- (e) It uses flash memory cells to store data.

10) Which of the following is/are **NOT** used as buses for communication between the CPU and main memory?

- (a) Data bus
- (b) Address bus
- (c) Arithmetic bus
- (d) Control bus
- (e) Interrupt bus

11) Which of the following highlight(s) the significance of the *Von Neumann architecture*?

- (a) Physically separate memory for data and instructions.
- (b) Ability to store programs and data in a single memory unit.
- (c) Integration of parallel processing as a core design principle.
- (d) Introduction of a structural model consisting of a CPU, Memory, and Input/Output.
- (e) The complete removal of data transfer bottlenecks.

12) Which of the following statements about printers is/are correct?

- (a) Inkjet printers are impact printers that print characters by striking an ink ribbon.
- (b) Dot matrix printers are commonly used for high-quality photo printing.
- (c) Laser printers rely on electrophotography and toner to produce printed output.
- (d) Thermal printers require liquid ink cartridges to operate.
- (e) Plotters are designed mainly for printing large volumes of plain text documents.

- 13) A full adder is implemented using a 3×8 decoder, where inputs A , B , and C_{in} are connected to the decoder inputs. Which of the following statements is/are correct?

Note: A *minterm* is a product term that represents one unique combination of the inputs A , B , and C_{in} for which the decoder asserts exactly one output line.

- (a) Each output of the 3×8 decoder corresponds to a unique *minterm* of A , B , and C_{in} .
- (b) The **Sum** output can be generated by connecting decoder outputs corresponding to *minterms* m_1 , m_2 , m_4 and m_7 through a OR gate.
- (c) The **Carry-out** output can be generated by performing logical OR on decoder outputs corresponding to *minterms* m_3 , m_5 , m_6 , and m_7 .
- (d) A 3×8 decoder alone is sufficient to generate both **Sum** and **Carry-out** without any additional logic gates.
- (e) A full adder cannot be implemented using a decoder.

- 14) Consider the following memory contents.

Address	Data Value
0x10	0x12
0x11	0x10
0x12	0x11

The instruction `LOAD R2, 0x10` loads a value into register R2 depending on the addressing mode used. What values will R2 contain under *immediate*, *direct*, and *indirect* addressing modes, respectively after executing the instruction?

- (a) 0x10, 0x11, 0x12
- (b) 0x12, 0x10, 0x11
- (c) 0x10, 0x12, 0x11
- (d) 0x11, 0x10, 0x12
- (e) 0x12, 0x11, 0x12

- 15) Which of the following input devices is/are primarily used for continuous or relative motion control, rather than discrete data entry?

- (a) Joystick
- (i) Stylus pen
- (c) Barcode reader
- (d) Touch pad
- (e) Trackball

- 16) Which of the following hardware devices provide(s) volatile storage?

- (a) Hard drive
- (b) Main Memory
- (c) CD-ROM
- (d) Flash memory stick
- (e) CPU Registers

17) Which of the following statements accurately describe(s) BIOS in a computer system?

- (a) BIOS initializes and tests hardware during boot-up.
- (b) BIOS manages internet connectivity and network hardware.
- (c) BIOS loads operating system from storage.
- (d) BIOS is hardware independent and all hardware use the same BIOS.
- (e) BIOS loads the basic device drivers.

18) Which of the following best explains the purpose of the beep codes during the Power-On-Self-Test (POST) process?

- (a) They provide diagnostic information when video output is not yet available.
- (b) They indicate errors that are detected only during the operating system loading.
- (c) They are used to configure BIOS settings automatically.
- (d) They confirm successful installation of application software.
- (e) They signal network connectivity issues during boot.

19) Which of the following activities is/are (an) example(s) of **passive preventive maintenance** in a computer system?

- (a) Periodically cleaning dust from internal components.
- (b) Updating the operating system and device drivers.
- (c) Using an uninterruptible power supply to protect against power failures.
- (d) Replacing a faulty cooling fan.
- (e) Running disk defragmentation utilities regularly.

20) Which of the following statements is/are correct?

- (a) OCR systems are used to convert scanned images of text into machine-readable and editable text.
- (b) A webcam typically captures images with higher DPI than a flatbed scanner.
- (c) OCR accuracy depends only on the webcam's frame rate (FPS).
- (d) A scanner with higher DPI produces more detailed digital images of printed documents.
- (e) A scanner is mainly designed for capturing real-time video streams.

21) What is/are the 8-bit **two's complement** representation for decimal value -126?

- | | | |
|---------------|---------------|---------------|
| (a) 0111 1110 | (b) 1000 0001 | (c) 1000 0010 |
| (d) 1111 1110 | (e) 1111 1101 | |

22) Which of the following statements regarding the SATA interface in computer systems is/are correct?

- (a) SATA utilizes parallel data transmission to achieve high-speed communication between the CPU and storage.
- (b) SATA is the standard interface for high-speed internal communication with storage devices such as HDDs and SSDs.
- (c) SATA technology is designed exclusively for external peripheral connectivity.
- (d) SATA uses a wide ribbon cable that allows multiple drives to share a single data path.
- (e) None of the above statements are correct.

23) Which of the following is a/are defining characteristic(s) of the traditional SCSI interface?

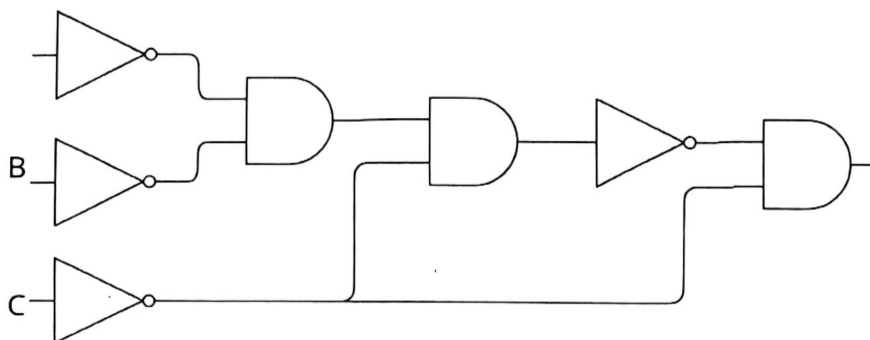
- (a) It is limited to connecting only a single internal device per controller.
- (b) It allows multiple internal and external devices on a single bus.
- (c) It was designed primarily to replace the USB standard.
- (d) It can be used to connect different types of devices such as scanners and printers.
- (e) It was designed exclusively for low-end home computers and is unsuitable for servers.

24) What is/are the simplified logical expression(s) for the following Karnaugh Map?

	$A \cdot \bar{B}$	$\bar{A} \cdot \bar{B}$	$\bar{A} \cdot B$	$A \cdot B$
C	1	0	0	1
$\bar{C}$	1	0	0	0

- (a) $\bar{A} \cdot \bar{B} + \bar{B} \cdot C$
- (b) $A \cdot \bar{B} + A \cdot B \cdot C$
- (c) $A \cdot (\bar{B} + C)$
- (d) $\bar{A} \cdot (B + \bar{C})$
- (e) $A \cdot \bar{B} + A \cdot C$

25) How many NAND gates are required to implement the given circuit, if all gates are replaced by their NAND-only equivalents and redundant double negations are removed?



- (a) 7
- (b) 8
- (c) 9
- (d) 10
- (e) 11

- 26) Consider a **synchronous 3-bit binary up-counter** implemented using **JK flip-flops**. All flip-flops share a common clock and the counter starts from **000**. Which of the following statements is/are correct?

- (a) The least significant flip-flop toggles on every clock pulse.
- (b) The most significant flip-flop toggles only when the two less significant flip-flops are both in the logic 1 state.
- (c) The counter reaches a maximum decimal value of 8, requiring 8 clock pulses to reach this state from the initial state 000.
- (d) After reaching the maximum positive number, the counter will start counting down to 000.
- (e) If the clock frequency is increased beyond the propagation delay limits of the flip-flops, the counter may produce incorrect counts.

- 27) Which of the following is/are the most critical requirement for any Real-Time Operating System?

- (a) User-friendly Interfaces
- (b) High Throughput
- (c) Determinism
- (d) Large Memory Footprint
- (e) Low Latency

- 28) Which of the following statements about device drivers is/are correct?

- (a) Device drivers are only necessary for external peripherals and are not used for internal components like hard drives.
- (b) Device drivers are typically both hardware-dependent and operating system-dependent.
- (c) Once a driver is installed, it replaces the BIOS/UEFI as the primary hardware initializer.
- (d) A single driver can be used for any hardware device as long as the operating system is the same.
- (e) Device drivers are stored permanently in the cache memory to ensure that the processor never has to wait for hardware interrupts.

- 29) Consider a RISC processor with a 5-stage pipeline, where each stage requires exactly t units of time to complete. If a sequence of 10 instructions is executed under ideal conditions (no pipeline stalls, hazards, or interrupts), what is the total time required to complete the execution of all 10 instructions?

- | | | |
|-----------|-----------|-----------|
| (a) $10t$ | (b) $14t$ | (c) $15t$ |
| (d) $45t$ | (e) $50t$ | |

30) If a processor's instruction set architecture (ISA) allocates a 3-bit field to specify a register operand, what is the maximum number of distinct general-purpose registers that can be directly addressed?

- | | | |
|--------|--------|--------|
| (a) 4 | (b) 8 | (c) 16 |
| (d) 32 | (e) 64 | |

31) Which of the following statements is/are **incorrect** regarding the Microsoft Disk Operating System?

- | |
|---|
| (a) It is a command-line based operating system where the user interacts with the system using text commands via a prompt. |
| (b) It is a single-user, single-tasking operating system, meaning only one user can operate it and only one program can run at a time. |
| (c) It supports a "Plug and Play" architecture that allows for the automatic configuration of complex hardware drivers without user intervention. |
| (d) It utilizes a 16-bit architecture and was originally designed to run on IBM PC compatible hardware. |
| (e) It manages files using a File Allocation Table (FAT) system. |

32) Which of the following statements correctly describe(s) fiber optic cable in computer networks?

- | |
|---|
| (a) A thick, shielded cable with a single copper wire in the center. |
| (b) A common network cable containing pairs of copper wires twisted together. |
| (c) A cable that uses electrical signals to transmit data through plastic wires. |
| (d) A common form of wired communication and are susceptible to electromagnetic interference. |
| (e) A glass-based cable used for high-speed internet over long distances. |

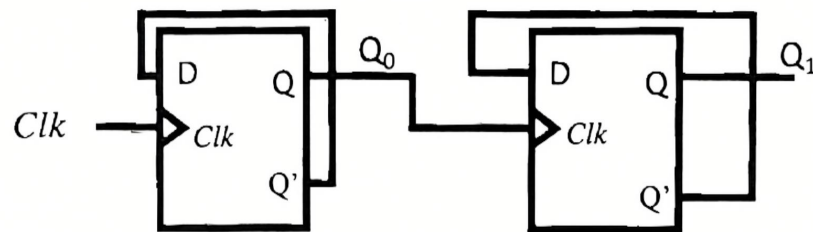
33) Which of the following statements correctly describe(s) a network router?

- | |
|--|
| (a) Operates only at the physical layer and forwards raw electrical signals. |
| (b) Fits in to one of the expansion slots of the motherboard of a computer. |
| (c) Is used mainly to convert digital signals into analog signals. |
| (d) Broadcasts data frames to all devices within the same network segment. |
| (e) Connects multiple networks and forwards packets based on IP addresses. |

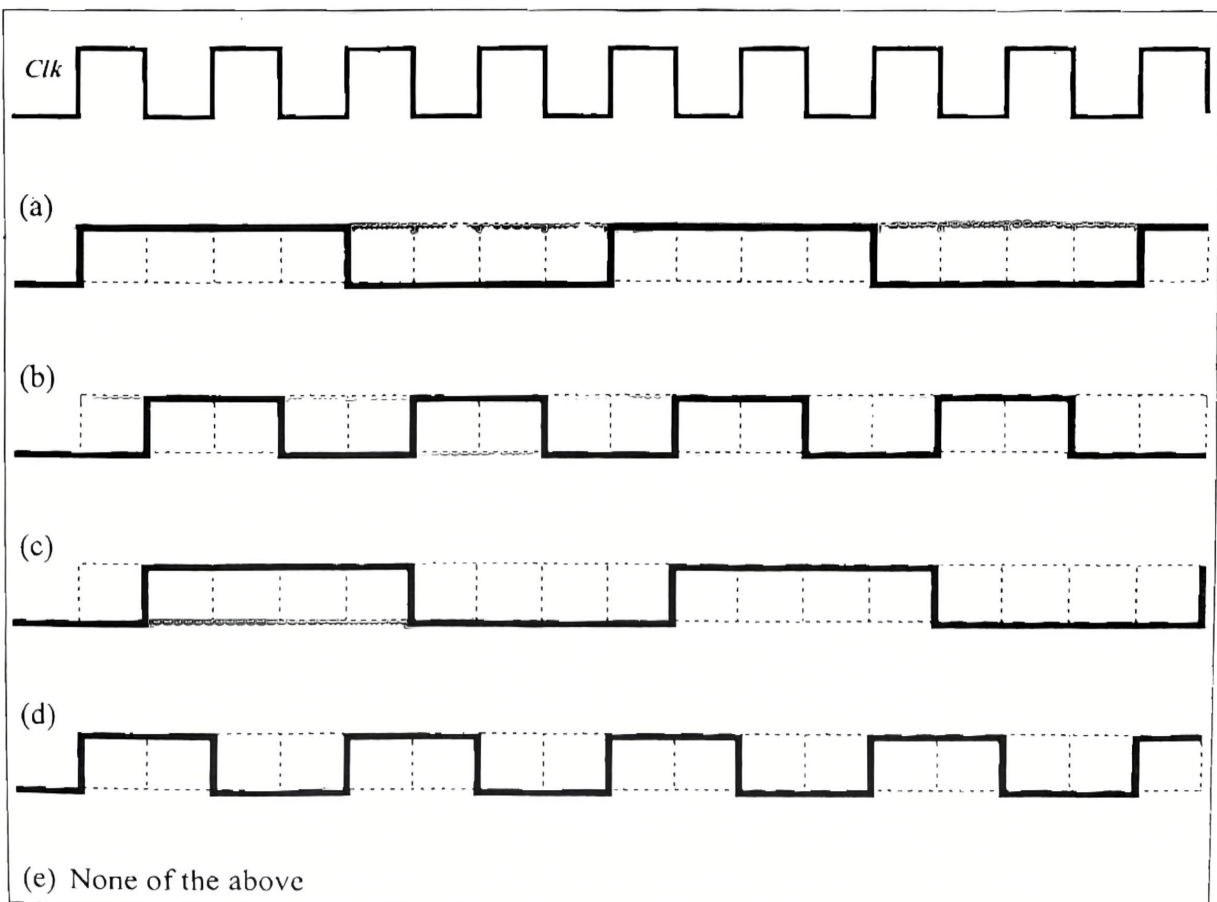
- 34) Which of the following is/are a/the simplified expression(s) of the Boolean algebraic expression $(\overline{A} \cdot \overline{B}) \cdot (A + B) \cdot (\overline{A} + B)$?

- | | |
|----------------------------|---------------------------------------|
| (a) $A \cdot \overline{B}$ | (b) $\overline{A} \cdot B$ |
| (c) $A + B$ | (d) $\overline{A} \cdot \overline{B}$ |
| (e) 0 | |

- 35) Consider the sequential logic circuit given below, which consists of two *positive edge triggered* D Flip Flops. The clock input is given as '*Clk*' in the diagram.



Assuming both Q_0 and Q_1 are logic value zero (0) at the beginning, which of the following time diagram(s) depict(s) the behaviour of Q_1 correctly for the given Clock (*Clk*) sequence?



36) Which of the following represents the mantissa (significand) component in the IEEE 754 single-precision representation of the decimal number 10.625?

- (a) 0101 0101 0000 0000 0000 0000
- (b) 101 0100 0000 0000 0000 0000
- (c) 010 1010, 0000 0000 0000 0000
- (d) 1010 1000 0000 0000 0000 0000
- (e) 0101 0100 0000 0000 0000 0000

37) Which of the following represents the exponent component in the IEEE 754 single-precision representation of the decimal number 10.625?

- | | | |
|---------------|---------------|---------------|
| (a) 0111 1110 | (b) 0000 0010 | (c) 0111 1101 |
| (d) 1000 0011 | (e) 1000 0010 | |

38) Which of the following Boolean expressions is equivalent to the expression $\overline{A + (B \cdot \overline{C})}$?

- | | | |
|---|---|---|
| (a) $\overline{A} \cdot \overline{B} + C$ | (b) $\overline{A} \cdot (\overline{B} + C)$ | (c) $\overline{A} + \overline{B} \cdot C$ |
| (d) $\overline{A} \cdot \overline{B} \cdot C$ | (e) $\overline{A} + \overline{B} + C$ | |

39) When the following machine instruction code is executed in a computer, what are the corresponding values stored in registers R2 and R3 respectively at the end of the final instruction execution?

Note: Interpretations of the machine instructions are as given here. All the numbers given in the machine code are in hexadecimal representation.

- **MOV R, A** Load the register R with hexadecimal value A.
- **ADD R1, R2, R3** Add the numeric values in R2 and R3 and place the results in R1.
- **INC R** Increases the value of register R by one.
- **BNE L, R, V** Branch to label L if the value in register R is Not Equal to value V.
- **LABEL :** Used as a jump target for branch instructions.

```
MOV R1, 0x0A
MOV R2, 0x00
LOOP:
ADD R1, R1, 0x05
INC R2
BNE LOOP, R2, 0x04
INC R2
INC R1
```

- | | | |
|---------------------|---------------------|---------------------|
| (a) R1 = 30, R2 = 4 | (b) R1 = 31, R2 = 5 | (c) R1 = 50, R2 = 4 |
| (d) R1 = 31, R2 = 4 | (e) R1 = 15, R2 = 5 | |

40) Which of the following statements correctly describe(s) the valid rules for grouping cells in a *Karnaugh Map* to simplify a Boolean expression?

- (a) Groups must be rectangular or square and can be formed horizontally, vertically, or diagonally.
- (b) Groups can wrap around the edges of the map, effectively treating the map as a torus (top-to-bottom and left-to-right).
- (c) The number of cells in any single group must be a power of 2 (e.g., 1, 2, 4, 8, or 16).
- (d) Every cell containing a '1' must be covered by at least one group to derive the complete simplified expression in sum of products form.
- (e) Overlapping groups is prohibited and often necessary to create the largest possible groups for maximum simplification.
